Research on Bi-parabolic Properties on Graphs

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Abstract

The research object of this paper is bi-parabolism of graphs. In section 1 and section 2, we firstly get an important gradients estimation inequality(theorem 2.2); And then in section 3 we sum up properties related to parabolism, such as the recurrence of Markov chain, and get similar conclusion of bi-parabolism with second-order difference; In the final section we explore the methods of using gradients estimation inequality to get Harnack inequality, and promote the conclusion to annulus.

Keywords: Markov chain; Gradients estimation inequality; Bi-parabolism

1 Definitions Related to Markov Chains and Motion on Graphs

Let $G(X, E)$ denote a graph or lattice consisting of a vertex set X and an edge set E (with $E \subset X \times X$). Throughout this paper, we assume by default that the graph is undirected and connected.

The study of motion on graphs has a long history. As early as 1959, Nash-Williams proposed methods for analyzing related Markov chains [8]. Soardi and Yamasaki [6], Fuchs and Zhong Kailai [7], Spitzer [9], and others conducted extensive research on random motion on graphs, including classification and applications. In 1975, Shing-Tung Yau and Shiu-Yuen Cheng established gradient inequalities on manifolds related to first-order differences [1, 5]. In 1995, Saloff-Coste applied these inequalities to graphs (the discrete case) and derived a proof of the Harnack inequality on lattices with at most quadratic growth [2].

For a random motion on G, we define the probability of moving from a point x to a point y in one step as $p(x, y)$, and the probability of reaching y

in n steps as $p^{(n)}(x,y)$. Let X be a finite or countable state space, and define the Markov chain (X,P) determined by X and the transition operator

$$P = (p(x,y))_{x,y\in X}.$$

Definition 1.1 (Irreducibility)**.** If for every $x, y \in X$, there exists some integer n such that

$$p^{(n)}(x,y) > 0,$$

then the Markov chain (X,P) is said to be **irreducible**.

Definition 1.2 (Recurrence and Transience)**.** If, starting from a certain state, the probability that the Markov chain returns to that state is equal to 1, then the chain is said to be **recurrent**. Conversely, if this probability is less than 1, the chain is said to be **transient**.

Definition 1.3 (Green's Function)**.** The Green's function associated with the Markov chain is defined by

$$G(x,y \mid z) = \sum_{n=0}^{\infty} p^{(n)}(x,y)\, z^n,$$

and in particular,

$$G(x,y) = G(x,y \mid 1).$$

By the Borel–Cantelli lemma, the Markov chain (X,P) is **recurrent** if and only if

$$G(x,y) = \infty \qquad \text{for all } x,y \in X.$$

Conversely, if this quantity is finite for some (and hence all) x, y, then (X,P) is **transient**.

Definition 1.4 (Superharmonic and Harmonic Functions)**.** For a function $f : X \to \mathbb{R}$, define

$$Pf(x) = \sum_{y\sim x} p(x,y) f(y),$$

where $y \sim x$ indicates that x and y are adjacent in the graph.

If

$$Pf(x) \le f(x) \qquad \text{for all } x \in X,$$

then f is called **superharmonic**. If

$$Pf(x) = f(x) \qquad \text{for all } x \in X,$$

then f is called **harmonic**.

Definition 1.5 (Reversibility)**.** A Markov chain (X, P) is said to be **reversible** if there exists a function (or measure)

$$m : X \to (0, \infty)$$

such that for all $x, y \in X$,

$$m(x)p(x, y) = m(y)p(y, x).$$

In this case, we define the quantity

$$\omega(x, y) = m(x)p(x, y) = m(y)p(y, x) = \omega(y, x)$$

to be the *conductance* of the edge (x, y), and $m(x)$ to be the *total conductance* at x.

Let $\rho(x, y)$ denote the graph distance between x and y (i.e., the number of edges in a shortest path). For $r \geq 0$, define the following local quantities:

$$W_x(r) = \sum_{\substack{z \in B_x(r),\, y \in X \\ \rho(z,x) < \rho(z,y)}} \omega(z, y),$$

$$V_x(r) = \sum_{z \in B_x(r),\, y \in X} \omega(z, y),$$

where $B_x(r)$ denotes the ball of radius r centered at x with respect to the metric ρ.

Conversely, if $\omega : X \times X \to [0, \infty)$ is symmetric and

$$m(x) = \sum_{y \in X} \omega(x, y)$$

is positive and finite for every $x \in X$, then the transition probability

$$p(x, y) = \frac{\omega(x, y)}{m(x)}$$

defines a reversible Markov chain (random walk) on X.

Definition 1.6. When $p = 2$, the (weighted) Laplacian of a function f at a vertex x is defined by

$$\Delta_2 f(x) = \sum_{y \sim x} \big(f(x) - f(y)\big)\, \omega(x, y).$$

The second-order Laplacian is defined by

$$\Delta_2^2 f(x) = \sum_{y\sim x} \big(\Delta_2 f(x) - \Delta_2 f(y)\big)\, \omega(x,y),$$

that is,

$$\Delta_2^2 f(x) = \sum_{y\sim x} \left\{ \sum_{x'\sim x} \big(f(x') - f(x)\big)\, \omega(x,x') - \sum_{y'\sim y} \big(f(y) - f(y')\big)\, \omega(y,y') \right\} \omega(x,y).$$

For a graph (or lattice), we consider the real Hilbert spaces of functions on the vertex and edge sets,

$$\ell^2(X,m) \qquad \text{and} \qquad \ell^2(E,r),$$

where X and E denote the vertex set and edge set, respectively. Their inner products are given by

$$(f,g) = \sum_{x\in X} f(x)\, g(x)\, m(x), \qquad \langle u,v\rangle = \sum_{e\in E} u(e)\, v(e)\, r(e),$$

where

$$r(e) = \frac{1}{\omega(e)}$$

is the *resistance* associated with the conductance $\omega(e)$ of the edge e.

Definition 1.7 (Difference Operator)**.** Define the difference operator

$$D : \ell^2(X,m) \longrightarrow \ell^2(E,r)$$

by

$$Df(e) = \frac{f(x) - f(y)}{r(e)},$$

where e is the edge connecting the vertices x and y.

The adjoint operator $D^* : \ell^2(E,r) \to \ell^2(X,m)$ is given by

$$D^* u(x) = \frac{1}{m(x)} \sum_{e\ni x} |u(e)|,$$

where the sum is taken over all edges incident to x.

The (weighted) Laplace operator is defined by

$$L = -D^*D = P - I,$$

where P is the transition operator of the associated random walk.

A direct computation yields

$$D^*(Df)(x) = \frac{1}{m(x)} \sum_{y \sim x} \frac{f(x) - f(y)}{r(x,y)}.$$

The quantity $m(x)D^*(Df)(x)$ corresponds to the sum of the terms appearing in the weighted Laplacian $\Delta f(x)$.

Considering the inner product $\langle Df, Df \rangle$ in $\ell^2(E, r)$, we obtain the *Dirichlet sum*

$$\langle Df, Df \rangle = \sum_{e=(x,y) \in E} \left(\frac{f(x) - f(y)}{r(e)} \right)^2 = \frac{1}{2} \sum_{x,y \in X} \big(f(x) - f(y)\big)^2 \omega(x,y),$$

where the factor $1/2$ accounts for symmetric double counting of edges.

Furthermore,

$$\langle \Delta_2 f, f \rangle = \frac{1}{2} \sum_{x \in X} |\nabla f(x)|^2,$$

where the gradient magnitude is defined by

$$|\nabla f(x)| = \left(\sum_{y \sim x} |f(x) - f(y)|^2 \, \omega(x,y) \right)^{1/2}.$$

For further details on random walks on graphs and the associated operators, see Wolfgang Woess[3], *Random Walks on Infinite Graphs and Groups.*

Lemma 1.8. *A function f is superharmonic if and only if $\Delta_2 f(x) \geq 0$ for all $x \in X$, and f is harmonic if and only if $\Delta_2 f(x) = 0$ for all $x \in X$.*

Proof. Recall that f is superharmonic if and only if $Pf \leq f$, where

$$Pf(x) = \sum_{y \sim x} p(x,y) f(y).$$

Multiplying both sides by $m(x)$ gives

$$m(x)Pf(x) = m(x)\sum_{y\sim x} p(x,y)f(y) = \sum_{y\sim x}\omega(x,y)f(y),$$

using $\omega(x,y) = m(x)p(x,y)$.

Similarly,

$$m(x)f(x) = \sum_{y\sim x} f(x)\omega(x,y).$$

Thus the inequality $Pf(x) \le f(x)$ is equivalent to

$$\sum_{y\sim x}\omega(x,y)f(y) \;\le\; \sum_{y\sim x}\omega(x,y)f(x),$$

which can be written as

$$\sum_{y\sim x}(f(x)-f(y))\,\omega(x,y) = \Delta_2 f(x) \ge 0.$$

Hence f is superharmonic if and only if $\Delta_2 f(x) \ge 0$. The case of equality characterizes harmonic functions. $\square$

2 Proof of Gradient Inequalities Related to Second-Order Differences

Definition 2.1. For any superharmonic function f defined on $B_x(R)$, define

$$\theta'_f(x,R) = \max_{\substack{z,y\in B_x(R)\\ z\sim y}} \frac{\Delta f(z)}{\Delta f(y)}$$

Lemma 2.2. *If f is superharmonic and $\Delta^2 f(x) > 0$, then*

$$\theta'_f(x,R) \leq \frac{1}{p(y,z)}$$

Proof. Since $\Delta^2 f(x) > 0$, we have

$$\sum_{\substack{y\sim x\\ \Delta f(x)>\Delta f(y)}} [\Delta f(x) - \Delta f(y)]\omega(x,y) \geq \sum_{\substack{y\sim x\\ \Delta f(x)>\Delta f(y)}} [\Delta f(y) - \Delta f(x)]\omega(x,y)$$

For any fixed $y \sim x$, they satisfy either $\Delta f(x) > \Delta f(y)$ or

$$\sum_{\substack{y\sim x\\ \Delta f(x)>\Delta f(y)}} [\Delta f(x) - \Delta f(y)]\omega(x,y) - \sum_{\substack{y\sim x\\ \Delta f(y)>\Delta f(x)\\ (x,y)\neq(x,y_0)}} [-(\Delta f(x) - \Delta f(y))]\omega(x,y)$$
$$\geq [\Delta f(y_0) - \Delta f(x)]\omega(x,y_0)$$

Since f is superharmonic, $\Delta f(x) > 0$, then

$$\sum_{\substack{y\sim x\\ \Delta f(x)>\Delta f(y)}} [\Delta f(x) - \Delta f(y)]\omega(x,y) - \sum_{\substack{y\sim x\\ \Delta f(y)>\Delta f(x)\\ (x,y)\neq(x,y_0)}} [-(\Delta f(x) - \Delta f(y))]\omega(x,y)$$
$$\leq \sum_{\substack{y\sim x\\ \Delta f(x)>\Delta f(y)}} \Delta f(x)\omega(x,y) + \sum_{\substack{y\sim x\\ \Delta f(y)>\Delta f(x)\\ (x,y)\neq(x,y_0)}} \Delta f(x)\omega(x,y)$$
$$= [m(x) - \omega(x,y_0)]\Delta f(x)$$

Thus, we can derive

$$(1 - p(x,y))\Delta f(x) \geq \Delta f(y)p(x,y) - \Delta f(x)p(x,y)$$

That is,

$$\frac{\Delta f(x)}{\Delta f(y)} \geq p(x,y)$$

Which means

$$\theta_f'(x,R) \leq \frac{1}{p(y,z)}$$

$\square$

Lemma 2.3. *For any $a \in (1,\infty)$ and $t \in \left[-1+\frac{1}{a},\, a-1\right]$, we have*

$$\big(-|\log(1+t)| + |t|\big)\,\mathrm{sgn}(t) \;\geq\; \frac{1}{8a^2}\,|\log(1+t)|^2.$$

Proof. Define

$$\psi_b(t) = \big(-|\log(1+t)| + |t|\big)\,\mathrm{sgn}(t) \;-\; \frac{b}{2}|\log(1+t)|^2.$$

Differentiating with respect to t, we obtain

$$\psi_b'(t) = \frac{1}{1+t}\big(t - b\;\mathrm{sgn}(t)\,|\log(1+t)|\big).$$

When $t > 0$, since $\mathrm{sgn}(t) = 1$,

$$\psi_b'(t) = \frac{1}{1+t}(t - b\log(1+t)).$$

For any $0 < b < 1$, we have $t \geq \log(1+t)$, hence $t - b\log(1+t) \geq 0$, so $\psi_b' \geq 0$. Thus ψ_b is increasing on $(0,\infty)$, and since $\psi_b(0) = 0$,

$$\psi_b(t) \geq 0, \qquad t > 0.$$

when $t \leq 0$, now $\mathrm{sgn}(t) = -1$, so

$$\psi_b'(t) = \frac{1}{1+t}\big(t - b\log(1+t)\big).$$

Let

$$g(t) = t - b\log(1+t).$$

Then

$$g'(t) = 1 - \frac{b}{1+t}.$$

Thus $g'(t) \geq 0$ whenever $t \in [-1+b, 0]$. Hence on this interval, g is increasing and $g(0) = 0$, so $g(t) \leq 0$. Thus $\psi_b'(t) \leq 0$ on $[-1+b, 0]$, so ψ_b is decreasing with $\psi_b(0) = 0$, hence

$$\psi_b(t) \geq 0, \qquad t \in [-1+b, 0].$$

Taking $b = 1/a$, we obtain for $t \in \left[-1 + \frac{1}{a}, \infty\right)$,

$$-|\log(1+t)| + |t| \;\geq\; \frac{1}{2a} |\log(1+t)|^2.$$

Since $a > 1$, we have

$$\frac{1}{2a} \geq \frac{1}{8a^2},$$

which yields the desired inequality.

$\square$

Lemma 2.4. *Let f be a superharmonic function defined on $B_x(R+1)$, and set $\theta' = \theta_f'(x, R+1)$. Then for all $z \in B_x(R)$,*

$$\Delta \log(\Delta f(z)) - \frac{\Delta^2 f(z)}{|\Delta f(z)|} \;\geq\; \frac{1}{8(\theta')^2} \left|\nabla \log \Delta f(z)\right|^2.$$

Proof. For every $y \sim z$, define

$$t'(y) = -1 + \frac{\Delta f(y)}{\Delta f(z)}.$$

Then

$$\log\left(\frac{\Delta f(z)}{\Delta f(y)}\right) = -\log(1 + t'(y)).$$

Thus

$$\begin{aligned}
&\Delta \log(\Delta f(z)) - \frac{\Delta^2 f(z)}{|\Delta f(z)|} \\
&= \sum_{y \sim z} \operatorname{sgn}\left(\log \frac{\Delta f(z)}{\Delta f(y)}\right) \left[\left|\log \frac{\Delta f(z)}{\Delta f(y)}\right| - \frac{|\Delta f(z) - \Delta f(y)|}{|\Delta f(z)|}\right] \omega(z, y) \\
&= \sum_{y \sim z} \operatorname{sgn}(t'(y)) \big(-|\log(1 + t'(y))| + |t'(y)|\big) \omega(z, y).
\end{aligned}$$

Because f is superharmonic and by definition of θ',

$$-1 + \frac{1}{\theta'} \;\le\; t'(y) \;\le\; \theta' - 1.$$

Applying Lemma 2.2 with $a = \theta'$, we obtain

$$\operatorname{sgn}(t'(y))\big(-|\log(1+t'(y))| + |t'(y)|\big) \ge \frac{1}{8(\theta')^2}|\log(1+t'(y))|^2.$$

Therefore,

$$\Delta \log(\Delta f(z)) - \frac{\Delta^2 f(z)}{|\Delta f(z)|} \;\ge\; \frac{1}{8(\theta')^2}\sum_{y\sim z}\big[\log \Delta f(y) - \log \Delta f(z)\big]^2 \omega(z,y).$$

Finally, recall that

$$|\nabla \log \Delta f(z)|^2 = \sum_{y\sim z}\big[\log \Delta f(y) - \log \Delta f(z)\big]^2 \omega(z,y).$$

Thus

$$\Delta \log(\Delta f(z)) - \frac{\Delta^2 f(z)}{|\Delta f(z)|} \;\ge\; \frac{1}{8(\theta')^2}\,|\nabla \log \Delta f(z)|^2.$$

$\square$

Theorem 2.5 ([1]). *Let M be a Riemannian manifold and let f be a superharmonic function defined on $B_0(R+1) \subset X$. For any increasing sequence of radii*

$$0 < r_0 < r_1 < \cdots < r_l = R,$$

we have

$$\int_{B(r_0)} |\nabla \log \Delta f(x)|^2 \;\le\; -8(\theta)^2 \int_{B(r_0)} \frac{\Delta^2 f(x)}{|\Delta f(x)|} + 64(\theta')^4 \left(\sum_{i=0}^{l-1} \frac{(r_{i+1}-r_i)^2}{W(r_{i+1}) - W(r_i)}\right)^{-1},$$

Proof. For $\delta > 0$, define a cutoff function

$$\phi_\delta(x) = \begin{cases} 0, & x \notin B(r), \\ \dfrac{1}{m(x)}, & x \in B(r-\delta), \end{cases}$$

extended smoothly on the annulus $B(r) \setminus B(r-\delta)$.

From Lemma 2.3 we have

$$\Delta \log\big(\Delta f(z)\big) - \frac{\Delta^2 f(z)}{|\Delta f(z)|} \;\geq\; \frac{1}{8(\theta')^2}\, |\nabla \log \Delta f(z)|^2.$$

Multiplying by ϕ_δ and integrating yields

$$\frac{1}{8(\theta')^2} \int_{B(r)} |\nabla \log \Delta f|^2\, \phi_\delta + \int_{B(r)} \frac{\Delta^2 f}{|\Delta f|} \;\leq\; \int_{B(r)\setminus B(r-\delta)} \phi_\delta\, \Delta \log(\Delta f).$$

Assuming $|\Delta \phi_\delta| \leq \delta^{-1}$, we estimate

$$\int_{B(r)\setminus B(r-\delta)} \phi_\delta\, \Delta \log(\Delta f) \;\leq\; \left(\frac{1}{\delta} \int_{B(r)\setminus B(r-\delta)} |\nabla \log \Delta f|^2\right)^{1/2} \left(\frac{1}{\delta}\big(W(r) - W(r-\delta)\big)\right)^{1/2}$$

Define

$$F(r) = \int_{B(r)} |\nabla \log \Delta f|^2, \qquad G(r) = -8(\theta')^2 \int_{B(r)} \frac{\Delta^2 f}{|\Delta f|}, \qquad c = \frac{1}{8(\theta')^2}.$$

Letting $\delta \to 0$, we obtain

$$c\,[F(r) - G(r)] \;\leq\; \sqrt{W'(r)}\,\sqrt{F'(r)}.$$

Hence,

$$\frac{1}{W'(r)} \;\leq\; \frac{1}{c^2} \frac{F'(r) - G'(r)}{(F(r) - G(r))^2} = -\frac{1}{c^2} \left(\frac{1}{(F(r) - G(r))^2}\right)'.$$

Integrating from r to r' gives

$$\frac{1}{c^2} \left(\frac{1}{F(r) - G(r)} - \frac{1}{F(r') - G(r')}\right) \;\geq\; \int_r^{r'} \frac{dx}{W'(x)}.$$

By Cauchy–Schwarz,

$$(r' - r)^2 \leq \left(\int_r^{r'} \frac{dx}{W'(x)}\right) \big(W(r') - W(r)\big),$$

so

$$\int_r^{r'} \frac{dx}{W'(x)} \;\geq\; \frac{(r' - r)^2}{W(r') - W(r)}.$$

Thus,

$$\frac{1}{F(r)-G(r)}-\frac{1}{F(r')-G(r')} \;\geq\; c^2\frac{(r'-r)^2}{W(r')-W(r)}.$$

For the sequence $r_0 < r_1 < \cdots < r_l = R$, summing over $i = 0, \ldots, l-1$ yields

$$\frac{1}{F(r_0)-G(r_0)}-\frac{1}{F(R)-G(R)} \;\geq\; c^2\sum_{i=0}^{l-1}\frac{(r_{i+1}-r_i)^2}{W(r_{i+1})-W(r_i)}.$$

Since $F(R) - G(R) > 0$, we have

$$F(r_0) \;\leq\; G(r_0)+\frac{1}{c^2}\left(\sum_{i=0}^{l-1}\frac{(r_{i+1}-r_i)^2}{W(r_{i+1})-W(r_i)}\right)^{-1}.$$

This completes the proof. $\square$

For the discrete (rather than continuous) case, we can derive an analogous estimate by employing similar techniques.

Theorem 2.6. *Let (G, ω) be a connected grid and let $p = 2$. Let f be a superharmonic function defined on $B_0(R+1) \subset X$. For any increasing sequence of positive integers*

$$r_0 < r_1 < \cdots < r_l = R,$$

we have

$$\sum_{B_0(r_0)}|\nabla \log \Delta f|^2 \leq 8\theta'^2\sum_{B_0(r_0)}\frac{-\Delta^2 f}{\Delta f}+64\theta'^4\left(\sum_{i=0}^{l-1}\frac{(r_{i+1}-r_i)^2}{W(r_{i+1})-W(r_i)}\right)^{-1} \tag{2.1}$$

Proof. Define $\phi(x)$ by

$$\phi(x)=\begin{cases}0, & x\notin B(r),\\ \dfrac{1}{m(x)}, & x\in B(r).\end{cases}$$

Taking the inner product with ϕ on both sides of the inequality in Lemma 2.3, we obtain

$$\frac{1}{8\theta'^2}\langle|\nabla\log\Delta f(z)|^2,\phi\rangle \le \left\langle \frac{-\Delta^2 f}{\Delta f} + |\log\Delta f(x)-\log\Delta f(y)|\,\mathrm{sgn}\left(\log\frac{\Delta f(x)}{\Delta f(y)}\right)\omega(x,y),\phi\right\rangle.$$

Thus

$$\begin{aligned}\frac{1}{8\theta'^2}\sum_{B(r)}|\nabla\log\Delta f(z)|^2 &\le \sum\frac{-\Delta^2 f}{\Delta f} + \sum_{\substack{x\sim y\\ x\in B(r)\\ y\notin B(r)}}|\log\Delta f(x)-\log\Delta f(y)|\,\mathrm{sgn}\left(\log\frac{\Delta f(x)}{\Delta f(y)}\right)\omega(x,y)\\ &\le \sum\frac{-\Delta^2 f}{\Delta f} + \big(W(r)-W(r-1)\big)^{1/2}\left(\sum_{B(r)\setminus B(r-1)}|\nabla\log\Delta f|^2\right)^{1/2}\end{aligned}$$

Using Hölder's inequality, set

$$F'(r)=\sum_{B(r)}|\nabla\log\Delta f|^2,\qquad G'(r)=8\theta'^2\sum_{B(r)}\frac{-\Delta^2 f}{\Delta f}.$$

Note that G is increasing. Let $K'(r)=F'(r)-G'(r)$. Then

$$K'(r)^2=\left[\sum_{B(r)}|\nabla\log\Delta f|^2+8\theta'^2\sum_{B(r)}\frac{\Delta^2 f}{\Delta f}\right]^2.$$

Next,

$$\frac{1}{8\theta'^2}\sum_{B(r)}|\nabla\log\Delta f|^2+\sum_{B(r)}\frac{\Delta^2 f}{\Delta f}\le\big(W(r)-W(r-1)\big)^{1/2}\left(\sum_{B(r)\setminus B(r-1)}|\nabla\log\Delta f|^2\right)^{1/2}.$$

Therefore,

$$K'(r)^2\le 64\theta'^4\big(W(r)-W(r-1)\big)\left(F'(r)-F'(r-1)\right).$$

Note that

$$F'(r)-F'(r-1)=\sum_{B(r)\setminus B(r-1)}|\nabla\log\Delta f|^2\le K'(r)-K'(r-1).$$

Hence,

$$\frac{1}{K'(r-1)} - \frac{1}{K'(r)} \geq \frac{K'(r) - K'(r-1)}{K'(r)^2} \geq \frac{1}{64\theta'^4\big(W(r) - W(r-1)\big)}.$$

For $0 < r < s \leq R$,

$$\frac{1}{K'(r)} - \frac{1}{K'(s)} \geq \frac{1}{64\theta'^4} \sum_{i=r+1}^{s} \frac{1}{W(i) - W(i-1)}.$$

Applying Hölder's inequality again:

$$\begin{aligned}
(s-r)^2 &= \left(\sum_{i=r+1}^{s} 1\right)^2 \\
&= \left(\sum_{i=r+1}^{s} [W(i) - W(i-1)]^{1/2} \frac{1}{[W(i) - W(i-1)]^{1/2}}\right)^2 \\
&\leq \left(\sum_{i=r+1}^{s} [W(i) - W(i-1)]\right) \left(\sum_{i=r+1}^{s} \frac{1}{W(i) - W(i-1)}\right) \\
&= \big(W(s) - W(r)\big) \left(\sum_{i=r+1}^{s} \frac{1}{W(i) - W(i-1)}\right).
\end{aligned}$$

Thus,

$$\frac{1}{K'(r-1)} - \frac{1}{K'(r)} \geq \frac{1}{64\theta'^4} \frac{(s-r)^2}{W(s) - W(r)}.$$

Consequently,

$$\frac{1}{K'(r_0)} \geq \frac{1}{64\theta'^4} \sum_{i=0}^{l-1} \frac{(r_{i+1} - r_i)^2}{W(r_{i+1}) - W(r_i)}.$$

Hence,

$$F'(r_0) - G'(r_0) \leq 64\theta'^4 \left(\sum_{i=0}^{l-1} \frac{(r_{i+1} - r_i)^2}{W(r_{i+1}) - W(r_i)}\right)^{-1},$$

which yields the desired inequality (2.1). □

3 Definitions of Parabolicity and Bi-parabolicity, Their Properties, and Connection with Markov Chains

We begin by introducing the functional space used throughout this section. Define the Banach space

$$D := \left\{ f : X \to \mathbb{R} \;\middle|\; \sum_{x \sim y} |f(x) - f(y)|^2 \omega(x,y) < \infty \right\},$$

equipped with the norm

$$N(f) = \left(|f(o)|^2 + \sum_{x \sim y} |f(x) - f(y)|^2 \, \omega(x,y) \right)^{1/2}.$$

Let D^0 be the closure of $l_0(X)$ in D, where $l_0(X)$ denotes the set of functions on X with finite support.

For the space $D = D(N)$ and fixed reference point o, the inner product may be written equivalently (see [3]) as

$$(f,g)_E = (f,g)_{E,o} = \langle Df, Dg \rangle + f(o)g(o),$$

where Df denotes the discrete gradient.

Lemma 3.1 ([3])**.** *The space D is a Hilbert space.*

Proof. Fix $x \in X$, $x \neq o$. Since the graph (X,E) is connected, there exists a path

$$o = x_0, x_1, \ldots, x_k = x, \qquad e_i = [x_{i-1}, x_i] \in E.$$

Let

$$C_1(x) = \sum_{i=1}^{k} r(e_i).$$

For $f \in D(N)$, using the Cauchy–Schwarz inequality,

$$\begin{aligned}
(f(x) - f(o))^2 &= \left(\sum_{i=1}^{k} \frac{f(x_i) - f(x_{i-1})}{\sqrt{r(e_i)}} \sqrt{r(e_i)} \right)^2 \\
&\leq \left(\sum_{i=1}^{k} \frac{(f(x_i) - f(x_{i-1}))^2}{r(e_i)} \right) \left(\sum_{i=1}^{k} r(e_i) \right) \\
&= \langle Df, Df \rangle \, C_1(x).
\end{aligned}$$

Define

$$C_2(x) = 2\max\{1,\, C_1(x)\}.$$

Then

$$f(x)^2 \le C_2(x)(f,f)_{E,o}.$$

Let $\{f_n\}$ be a Cauchy sequence in $D(N)$. Then for every fixed $x \in X$, the sequence $\{f_n(x)\}$ is Cauchy in $\mathbb{R}$, hence converges to some finite $f(x)$. Moreover, $\{Df_n\}$ is Cauchy in $l^2(E,r)$, so there exists u such that $Df_n \to u$. By definition of the gradient, $u = Df$ and $Df < \infty$.

Furthermore, the above estimate shows that convergence in $D(N)$ implies pointwise convergence.

Thus D is complete under the inner product $(\cdot,\cdot)_{E,o}$, hence a Hilbert space. □

We now recall a standard property of Hilbert spaces (see [11]):

Lemma 3.2. *Let H be a Hilbert space, and let $E \subset H$ be a non-empty closed convex set. Then there exists a unique element $x_0 \in E$ of minimal norm:*

$$\|x_0\| \le \|x\| \quad \forall x \in E.$$

Proof. Let

$$p = \inf\{\|x\| : x \in E\}.$$

Using the parallelogram identity,

$$\|x+y\|^2 + \|x-y\|^2 = 2\|x\|^2 + 2\|y\|^2,$$

we obtain

$$\frac{1}{4}\|x-y\|^2 = \frac{1}{2}\|x\|^2 + \frac{1}{2}\|y\|^2 - \left\|\frac{x+y}{2}\right\|^2.$$

Since E is convex, $\frac{x+y}{2} \in E$, and thus

$$\|x-y\|^2 \le 2\|x\|^2 + 2\|y\|^2 - 4p^2.$$

If $\|x\| = \|y\| = p$, then the above implies $\|x-y\| = 0$, hence $x = y$. Thus the minimizer is unique.

Let $\{x_n\} \subset E$ satisfy $\|x_n\| \to p$. Applying the above inequality to x_n and x_m, the right-hand side tends to 0 as $m, n \to \infty$. Thus $\{x_n\}$ is Cauchy, and converges to some x_0. Since E is closed, $x_0 \in E$, and continuity of the norm gives

$$\|x_0\| = \lim_{n\to\infty} \|x_n\| = p.$$

□

If $A \subset X$, we denote by P_A the restriction of the operator P to the set A.

$$P_A(x,y) = \begin{cases} p(x,y), & x, y \in A, \\ 0, & \text{otherwise}. \end{cases} \tag{3.1}$$

The corresponding Green function is defined by

$$G_A(x, y \mid z) = \sum_{n=0}^{\infty} p_A^{(n)}(x,y) z^n.$$

If A is finite then $G_A < \infty$, and we have

$$(I_A - P_A) G_A = I_A.$$

If $f \in l_0(X)$ and $\operatorname{supp}(f) \subset A$, then

$$\langle Df,\, DG_A(\cdot, x)\rangle = (f, (I-P)G_A(\cdot,x)) = (f, (I_A-P_A)G_A(\cdot,x)) = (f, \delta_x) = m(x)f(x).$$

Definition 3.3 (Parabolicity)**.** If

$$\inf \left\{ \sum_{x \sim y} |f(x) - f(y)|^2\, \omega(x,y) \;\middle|\; f \in l_0(X),\ f(o) = 1 \right\} = 0,$$

then the graph is called *parabolic*.

From the above functional-analytic considerations, we also have

$$\sum_{x \sim y} |f(x) - f(y)|^2 \omega(x,y) = 2\langle Df, Df\rangle.$$

Theorem 3.4 (Equivalent characterizations of recurrence [9])**.** *The following statements are equivalent:*

(1) *The Markov chain (X, P) is recurrent,*
(2) *The graph is parabolic,*
(3) $\mathbf{1} \in D^0(N)$,
(4) *Every nonnegative superharmonic function is constant.*

Proof. **(a) ⇔ (d).** If (X, P) is transient, then for any $y \in X$, the function $x \mapsto G(x, y)$ is superharmonic but not harmonic, hence nonconstant.

Conversely, if (X, P) is recurrent and f is nonnegative superharmonic, set $g = f - Pf$. If $g(y_0) > 0$ for some y_0, then for all n,

$$\sum_{k=0}^{n} p^{(k)}(x, y_0)\, g(y_0) \leq f(x) - P^{n+1} f(x) \leq f(x).$$

Thus $G(x, y_0) \leq f(x)/g(y_0)$, contradicting recurrence. Hence every nonnegative superharmonic function must be harmonic. Fix x_0, let $M = f(x_0)$, and set $h(x) = \min\{M, f(x)\}$; then h is harmonic. By the maximum principle, $h \equiv M$, hence f is constant.

(a) ⇒ (b). If the graph is transient, define

$$u = -\frac{a_0}{m(x)}\, DG(\cdot, x).$$

Then

$$D^* u = \frac{a_0}{m(x)} LG(\cdot, x) = -\frac{a_0}{m(x)} \delta_x.$$

Taking $a_0 = -1$, we have

$$\langle Df, u\rangle = (f, D^* u) = \left(f, \frac{1}{m(x)} \delta_x\right) = f(x) = 1.$$

Hence

$$1 = |\langle Df, u\rangle|^2 \leq \langle Df, Df\rangle\, \langle u, u\rangle,$$

so

$$\langle Df, Df\rangle \geq \frac{1}{\langle u, u\rangle} > 0.$$

(b) ⇒ (a). Let $A \subset X$ finite, $x \in A$, and set

$$f = \frac{G_A(\cdot, x)}{G_A(x, x)}.$$

Then $f \in D^0$ and $f(x) = 1$. Using previous identities,

$$\sum |f(x) - f(y)|^2 \omega(x, y) = \langle Df, Df\rangle = \frac{1}{G_A(x, x)^2} \langle DG_A(\cdot, x), DG_A(\cdot, x)\rangle = \frac{m(x)}{G_A(x, x)}.$$

If the graph is non-parabolic, then

$$G_A(x,x) \le \frac{m(x)}{\inf \sum |f(x)-f(y)|^2 \omega(x,y)},$$

so

$$G(x,x) < \infty,$$

hence the chain is transient. If the graph is parabolic then $G(x,x)=\infty$, i.e. the chain is recurrent.

(b) $\Leftrightarrow$ (c). If the graph is parabolic, there exists a Cauchy sequence f_n with $N(f_n) \to 0$. In the Hilbert space D, $f_n \to f_0 \in D^0$, and since each $f_n(o) = 1$, we have $f_0(o) = 1$. Thus $N(f_0) = 0$, so for any edge $\{x, y\}$, $f_0(x) = f_0(y) = 1$, hence $f_0 \equiv 1 \in D^0$.

Conversely, suppose $1 \equiv f_0 \in D^0$. Let $F(e) = |f(x) - f(y)|$ for the two endpoints of edge e. Consider

$$\sum |F(e) + tG(e)|^2 \omega(e),$$

where $G(e) = |g(x) - g(y)|$ and $g(o) = 0$. Differentiating in t,

$$\sum 2[F(e) + tG(e)]G(e)\,\omega(e).$$

If f achieves the infimum, then at $t = 0$,

$$\sum F(e)G(e)\,\omega(e) = 0.$$

Choosing $G(e) = 1$ for all e gives

$$\sum F(e)\,\omega(e) = 0.$$

Since $\omega(e) > 0$, we obtain

$$\inf \sum |f(x) - f(y)|^2 \omega(x,y) = 0,$$

hence the graph is parabolic. □

Definition 3.5 (Excessive and invariant measures)**.** Let $v : X \to [0,\infty)$ be a function such that

$$vP(y) = \sum_{x \in X} v(x)p(x,y)$$

where the sum is assumed finite. If $vP \le v$ pointwise, we call v an *excessive function* (or measure). If $vP = v$ pointwise, we call v an *invariant function* (or measure).

Using this definition, we obtain another characterization of recurrence, parallel to the equivalence between recurrence and constancy of superharmonic functions.

Corollary 3.6. *The Markov chain (X, P) is recurrent if and only if there exists an invariant measure v such that every excessive measure is a scalar multiple of v.*

Proof. (⇒) Assume (X, P) is recurrent. For any excessive measure v, the inequality

$$vP(x) = \sum_{y \sim x} v(y)p(x, y) \leq v(x)$$

shows that v is superharmonic. By recurrence, every nonnegative superharmonic function is constant; hence $v \equiv c$ for some $c > 0$.

Now consider the function $v_0 \equiv 1$. Since

$$v_0 P(x) = \sum_{y \sim x} p(x, y) = 1,$$

the function v_0 is an invariant measure. Thus

$$v = cv_0.$$

(⇐) Conversely, suppose every excessive measure is a multiple of some invariant measure v_0. Let f be any superharmonic function. Then

$$Pf(x) = \sum_{y \sim x} p(x, y)f(y) \leq f(x),$$

so f is excessive. By assumption,

$$f = c_0 v_0$$

for some $c_0 \geq 0$.

To conclude recurrence using Theorem 3.1(d), it suffices to show that v_0 is constant. Suppose, toward a contradiction, that v_0 is not constant. Because X is countable, there exists a point $x_1 \in X$ such that

$$v_0(x_1) = \min_{x \in X} v_0(x),$$

and some x_2 with

$$v_0(x_2) > v_0(x_1).$$

If no neighbor of x_1 has a different value from $v_0(x_1)$, replace x_1 by a neighboring point and iterate; since the graph is connected, we must eventually reach adjacent vertices with different values. Thus we may assume $x_2 \sim x_1$ and $v_0(x_2) > v_0(x_1)$. Then

$$v_0 P(x_1) = \sum_{y \sim x_1} v_0(y) p(x_1, y) > v_0(x_1) \sum_{y \sim x_1} p(x_1, y) = v_0(x_1),$$

contradicting the invariance $v_0 P = v_0$. Therefore v_0 must be constant, and hence f is constant.

By Theorem 3.1(d), the chain (X, P) is recurrent. □

A more "physical" interpretation of recurrence is given through the notion of finite-energy flow[3].

Definition 3.7 (Finite-energy flow)**.** Given a point x_0 and a real number a_0, a *finite-energy flow* from x_0 to infinity with input energy a_0 is a function $u \in \ell^2(E, r)$ satisfying

$$\nabla^* u(y) = -\frac{a_0}{m(x_0)}\, \delta_{x_0}(y), \qquad y \in X.$$

Its *energy* is defined as $\langle u, u \rangle$.

The relation between recurrence of a Markov chain and the "energy" required to reach infinity is expressed by

$$\frac{G(x,x)}{m(x)} = \frac{1}{\inf_f \sum_{(x,y) \in E} |f(x) - f(y)|^2 \omega(x,y)}.$$

Thus parabolicity is equivalent to the energy required to escape to infinity being infinite, which corresponds to recurrence.

For the case of p-parabolicity, analogous equivalences hold:

$$\begin{aligned} (G,\omega) \text{ is } p\text{-parabolic} &\iff 1 \in D_0(N) \\ &\iff \text{all nonnegative } p\text{-superharmonic functions are constant.} \end{aligned}$$

However, the connection with Markov chain recurrence applies only for $p = 2$.

Between the property $1 \in D_0(N)$ and recurrence lies an important criterion.

Theorem 3.8 (Nash–Williams Criterion [8])**.** *Suppose that X admits a partition*

$$X = \bigcup_{i\in I} X_i, \qquad \textit{with} \quad \mathbf{1}_{X_i} \in D_0(N) \ \ \textit{for all } i \in I.$$

Define

$$\omega'(i,j) = \begin{cases} \displaystyle\sum_{\substack{x\in X_i\\ y\in X_j}} \omega(x,y), & i \neq j, \\ 0, & i = j. \end{cases}$$

Assume $I = \mathbb{N}_0$ and $\omega'(i,j) = 0$ whenever $|i-j| \geq 2$. If

$$\sum_{i=1}^{\infty} \frac{1}{\omega'(i-1,i)} = \infty,$$

then the Markov chain (X,P) is recurrent.

Now let us go to bi-parabolicity and second-order energy:

Definition 3.9 (Second-order energy)**.** Define

$$E_2(f) = \sum_{(x,y)\in E} |\Delta f(x) - \Delta f(y)|^2 \omega(x,y) = 2\langle D(\Delta f), D(\Delta f)\rangle.$$

Let

$$E_2 = \{\Delta f : E_2(f) < \infty\}, \qquad E_2^0 = \overline{\{\Delta f : \Delta f \text{ has finite support}\}}^{E_2}.$$

By Lemma 3.2, for any fixed $c \in \mathbb{R}$,

$$\inf\{E_2(f) : \Delta f \in \ell_0(X),\ \Delta f(o) = c\} = \min\left\{E_2(f) : f \in E_2^0,\ \Delta f(o) = c\right\}.$$

Let $f_0 \in E_2^0$ realize the minimum with $\Delta f_0(o) = c$. Then

$$\inf\{E_2(f) : \Delta f(o) = c\} = E_2(f_0)$$

Definition 3.10 (Bi-parabolicity)**.** If

$$\inf\left\{\sum_{(x,y)} |\Delta f(x) - \Delta f(y)|^2\, \omega(x,y) : \Delta f \in \ell_0(X),\ \Delta f(o) = c\right\} = 0,$$

(i.e. Δf has finite support, or equivalently f is 2-superharmonic on a compact set and 2-harmonic outside), then the grid (G,ω) is called *bi-parabolic.*

Bi-parabolicity implies that the minimizer satisfies

$$E_2(f_0) = 0, \qquad \text{so} \qquad \Delta f_0(x) = \Delta f_0(y) = \Delta f_0(o) = c \quad \forall x, y \in X.$$

Furthermore,

$$\Delta^2 f_0(x) = 0 \qquad \text{for } x \neq o.$$

Hence, if every function satisfying $\Delta^2 f \geq 0$ has $\Delta f \equiv c$, then the grid satisfies $\inf \{E_2(f), \Delta f \in l_0(X), \Delta f(o) = c\} = 0$.

Corollary 3.11. *For a grid (G, ω), suppose there exist a point o and an infinite sequence $r_0 < r_1 < \cdots < r_i < \cdots$ such that*

$$\sum_{i=0}^{\infty} \frac{(r_{i+1} - r_i)^2}{W(r_{i+1}) - W(r_i)} = +\infty.$$

Then

$$\inf \left\{ \sum_{(x,y)\in E} |\Delta f(x) - \Delta f(y)|^2 \, \omega(x,y) : \Delta f \in \ell_0(X), \ \Delta f(o) = c \right\} = 0,$$

i.e. the grid is bi-parabolic.

Proof. Let f satisfy $\Delta^2 f \geq 0$. If $\theta'_f(0, \infty) < \infty$, then by the gradient inequality in Theorem 2.2,

$$\Delta f \equiv c.$$

If instead $\theta'_f(0, \infty) = \infty$, define

$$\Delta f_t := \min\{1 + \Delta f, \, t\}, \qquad t > 0.$$

Then $\theta'_{f_t} \leq t$, hence Δf_t is constant. As this holds for all t, letting $t \to \infty$ yields $\Delta f \equiv c$. Thus

$$\inf\{E_2(f) : \Delta f(o) = c\} = 0.$$

We now show that for any grid (G, ω), if $\Delta f \equiv c$, then $c = 0$. Equivalently, the system

$$\Delta f(x) = c, \qquad x \in X,$$

admits no solution with $c \neq 0$. Letting $\omega(x, y) = 0$ whenever x and y are not adjacent, we write explicitly:

$$\begin{aligned}
\Delta f(x_1) &= c = \sum_{y \sim x_1} (f(x_1) - f(y))\, \omega(x_1, y) \\
&= \omega(x_1, x_2)\big(f(x_1) - f(x_2)\big) + \cdots + \omega(x_1, x_n)\big(f(x_1) - f(x_n)\big), \\
\Delta f(x_2) &= c = \omega(x_2, x_1)\big(f(x_2) - f(x_1)\big) + \cdots + \omega(x_2, x_n)\big(f(x_2) - f(x_n)\big) \\
&= -\omega(x_1, x_2)\big(f(x_1) - f(x_2)\big) + \cdots, \\
&\vdots \\
\Delta f(x_n) &= c = \omega(x_n, x_1)\big(f(x_n) - f(x_1)\big) + \cdots + \omega(x_n, x_{n-1})\big(f(x_n) - f(x_{n-1})\big) \\
&= -\omega(x_1, x_n)\big(f(x_1) - f(x_n)\big) + \cdots .
\end{aligned}$$

For a grid with n vertices, the unknowns are the pairwise differences

$$f(x_i) - f(x_j), \qquad 1 \leq i < j \leq n,$$

totalling $\binom{n}{2}$ variables, while the system consists of n equations. The associated coefficient matrix takes the form

$$\begin{pmatrix}
\omega(x_1,x_2) & \omega(x_1,x_3) & \cdots & \cdots & \omega(x_1,x_n) & \cdots & 0 & \cdots & 0 & \\
-\omega(x_1,x_2) & 0 & \cdots & 0 & \cdots & \omega(x_2,x_3) & \cdots & \omega(x_2,x_n) & \cdots & 0 \\
0 & -\omega(x_1,x_3) & 0 & \cdots & 0 & -\omega(x_2,x_3) & \cdots & 0 & \cdots & 0 \\
\vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\
0 & 0 & \cdots & \cdots & -\omega(x_n,x_1) & 0 & \cdots & -\omega(x_2,x_n) & \cdots & -\omega(x_{n-1},x_n)
\end{pmatrix}$$

Denote the row vectors by $\alpha_1, \ldots, \alpha_n$. Each row has at most n nonzero entries; each column has exactly two nonzero entries of opposite sign, and no two such entries lie in the same row. Summing all rows yields

$$\alpha_1 + \cdots + \alpha_n \equiv 0.$$

Summing the right-hand side of the system gives

$$nc = \alpha_1 + \cdots + \alpha_n \equiv 0,$$

hence $c = 0$.

Therefore $\Delta f \equiv c$ implies $c = 0$, and so $\Delta f \equiv 0$, i.e. $f \equiv c_0$. This completes the proof. $\square$

4 Corollaries on Harnack Inequality and Extensions of Previous Gradient Inequalities to Annular Regions

Definition 4.1 (Connectivity Property)**.** Given $o \in X$ and positive integers s, R with $s < \infty$, a graph G is said to satisfy property $[C(o, s, R)]$ if for all $0 \le r \le R - s$, the annulus

$$B_0(r+s) \setminus B_0(r)$$

is connected.

From Definition 4.1 we see that:

$\mathbb{Z}^d$ satisfies $[C(o, 2, \infty)]$ but not $[C(o, 1, \infty)]$, $\qquad \mathbb{R}^d$ satisfies $[C(o, \varepsilon, \infty)]$ for all $\varepsilon > 0$.

Theorem 4.2 (Discrete Harnack Inequality)**.** *Let (G, ω) be a connected weighted graph and assume*

$$\overline{\omega} := \sup_{e \in E} \frac{1}{\omega(e)} < \infty.$$

Fix $o \in X$ and integers s, T, R satisfying $0 < 2s \le T \le R$. If G satisfies $[C(o, s, R+T)]$, then for any function f with $\Delta^2 f = 0$ on $B_0(R+T+1)$, we have

$$\sup_{B_0(R+1)} \Delta f \;\le\; \exp\!\left(32 s^2\, \overline{\omega}\, \theta'^2\, \frac{V(2R)}{T^2} \right) \inf_{B_0(R+1)} \Delta f,$$

where $\theta' = \theta'_f(o, R+T+1)$ and $V(r) = V_0(r)$.

Proof. Write $B_0(r) = B(r)$ and $V_0(r) = V(r)$. Set

$$t = \left\lfloor \tfrac{T}{2s} \right\rfloor.$$

By Hölder's inequality,

$$\left(\sum_{B(R+t)} \nabla_1 \log \Delta f \right)^2 \le V(R+t) \sum_{B(R+t)} |\nabla \log \Delta f|^2. \tag{4.1}$$

From the gradient bound in Theorem 2.2,

$$\sum_{B_0(r_0)} |\nabla \log \Delta f|^2 \le 64 \theta'^4 \left(\sum_{i=0}^{\ell-1} \frac{(r_{i+1} - r_i)^2}{W(r_{i+1}) - W(r_i)} \right)^{-1}. \tag{4.2}$$

Choose $r_0 = R+t$, $\ell = 1$, $r_1 = R+T$. Using $R+T \leq 2R$ and $T-t \geq T/2$, inequality (4.2) yields

$$V(R+t) \sum_{B(R+t)} |\nabla \log \Delta f|^2 \leq 64\theta'^4 \frac{4V(2R)^2}{T^2}. \tag{4.3}$$

Define annuli

$$\Sigma_i := B\big(R+(i+1)s\big) \setminus B\big(R+is\big), \qquad 0 \leq i \leq t/s - 1.$$

From (4.3) we obtain

$$\sum_{i=0}^{t/s-1} \sum_{\Sigma_i} \nabla_1 \log \Delta f \leq 8\theta'^2 \frac{2V(2R)}{T}. \tag{4.4}$$

Therefore, for some $j \in \{0, \ldots, t/s - 1\}$,

$$\frac{t}{s} \sum_{\Sigma_j} \nabla_1 \log \Delta f \leq 8\theta'^2 \frac{2V(2R)}{T}. \tag{4.5}$$

Let $X_M, X_m \in \Sigma_j$ be points where Δf achieves respectively its maximum and minimum on Σ_j. By the maximum principle,

$$\frac{\sup_{B(R+1)} \Delta f}{\inf_{B(R+1)} \Delta f} \leq \frac{\Delta f(X_M)}{\Delta f(X_m)}. \tag{4.6}$$

Since Σ_j is connected by assumption $[C(o, s, R+T)]$, we have

$$\begin{aligned} \log\left(\frac{\Delta f(X_M)}{\Delta f(X_m)}\right) &\leq \sum_{\substack{e=(x,y) \\ x,y \in \Sigma_j}} |\log \Delta f(x) - \log \Delta f(y)| \\ &\leq \overline{\omega} \sum_{\substack{e=(x,y) \\ x,y \in \Sigma_j}} |\log \Delta f(x) - \log \Delta f(y)|\, \omega(x,y) \qquad (4.7) \\ &= \overline{\omega} \sum_{\Sigma_j} \nabla_1 \log \Delta f. \end{aligned}$$

Combining with (4.5),

$$\log\left(\frac{\Delta f(X_M)}{\Delta f(X_m)}\right) \leq \overline{\omega} \cdot 8\theta'^2 \frac{2sV(2R)}{T\,\lfloor T/(2s) \rfloor}. \tag{4.8}$$

Exponentiating,

$$\frac{\sup_{B(R+1)} \Delta f}{\inf_{B(R+1)} \Delta f} \leq \exp\left(\overline{\omega} \cdot 8\theta'^2 \frac{2sV(2R)}{T\lfloor T/(2s) \rfloor}\right) \leq \exp\left(32s^2 \overline{\omega} \theta'^2 \frac{V(2R)}{T^2}\right),$$

completing the proof. $\square$

From the definition we note that $\mathbb{Z}^d$ satisfies $[C(o, 2, \infty)]$ but not $[C(o, 1, \infty)]$, while $\mathbb{R}^d$ satisfies $[C(o, \varepsilon, \infty)]$ for any $\varepsilon > 0$.

Corollary 4.3. *Let (G, ω) be a connected grid, and let*

$$\overline{\omega} = \sup_{e \in E} \frac{1}{\omega(e)} < \infty.$$

Fix a point $o \in X$ and $s \geq 1$, and assume that G satisfies $[C(o, s, \infty)]$ and

$$\sup_{r \geq 1} \{r^{-2} V_o(r)\} = Q < \infty.$$

Then for all $R \geq 2$ and all functions f satisfying $\Delta^2 f \geq 0$ on $B_0(2R+1)$, we have

$$\sup_{B_0(R+1)} \Delta f \ \leq \ e^{128 s^2 \overline{\omega}^2 \theta'^2 / 2} \inf_{B_0(R+1)} \Delta f,$$

where $\theta' = \theta'_f(o, 2R+1)$.

In particular, if we further assume

$$m = \sup_{E} \frac{1}{p(e)} < \infty,$$

then for any $R \geq 2$ and any positive 2-harmonic function f on $B_0(2R+1)$, we have

$$\sup_{B_0(R+1)} \Delta f \ \leq \ C \inf_{B_0(R+1)} \Delta f,$$

where

$$C = 128 s^2 \overline{\omega}\, m^2 Q.$$

Corollary 4.4. *Let (G, ω) be a connected grid, and let*

$$\overline{\omega} = \sup_{e \in E} \frac{1}{\omega(e)} < \infty.$$

Let
$$m = \sup_E \frac{1}{p(e)} < \infty,$$
and fix a point $o \in X$ and $s \geq 1$. Assume that G satisfies $[C(o, s, \infty)]$.
$$\sup_{r \geq 1} \{r^{-2} V_0(r)\} = Q < \infty.$$

Then there exist $\epsilon > 0$ and K such that for all functions f satisfying $\Delta^2 f \geq 0$ on $B_0(2R+1)$, we have
$$\sup_{x,y \in B_0(r)} \big|\Delta f(x) - \Delta f(y)\big| \ \leq \ K(r/R)^\epsilon \inf_{B_0(R+1)} \Delta f, \qquad \forall\, 0 < r \leq R.$$

Actually, by subtracting 1 from the coefficient on the right-hand side of the above formula, we can obtain this corollary.

Now we extend the conclusions into annulus: For $0 \leq r < R \leq \infty$, define the annulus
$$A_o(r, R) = B_o(R) \setminus B_o(r),$$
and set
$$U_o^{r,R}(s) = \big[W_o(r) - W_o(r-s)\big] + \big[W_o(R+s) - W_o(R)\big].$$

Let $1 \leq r \leq R$ and assume f satisfies $\Delta^2 f \geq 0$ on $A_o(r-1, R+1)$. Define
$$\theta' = \sup\left\{\log \frac{\Delta f(x)}{\Delta f(y)} : \ x, y \in A_o(r-1, R+1), \ x \sim y\right\}.$$

Then, from Lemma 2.3 and applying Hölder's inequality, we can similarly obtain
$$\frac{1}{8\theta^2} \sum_{A(s,t)} |\nabla_2 \log \Delta f|^2 \leq \sum_{A(s,t)} \frac{-\Delta_2^2 f}{\Delta f} + [U^{s,t}(1)]^{1/2} (\sum_{A'(s,t)} |\nabla_2 \log \Delta f|^2)^{1/2}$$

where, $A'(s,t) = A(s,t) \setminus A(s+1, t-1)$. From this, using similar reasoning and steps, we can derive:

Theorem 4.5 (Gradient Estimate on Annuli)**.** *Let (G,ω) be a connected grid and let f satisfy $\Delta^2 f \geq 0$ on $A_o(r-1, R+1)$. If $r < s < t < R$ and $0 = n_0 < n_1 < \cdots < n_l = \min\{R-t,\, s-r\}$, then*

$$\sum_{A_o(s,t)} |\nabla \log \Delta f|^2 \leq 64\,\theta'^4 \left(\sum_{i=0}^{l-1} \frac{(n_{i+1}-n_i)^2}{U(n_{i+1}) - U(n_i)} \right)^{-1}.$$

Corollary 4.6. *Setting $s = 2R$, $t = 5R$, $l = 1$, $n_1 = R$, and noting that*

$$U_o^{2R,5R}(1) = W_o(2R) - W_o(R) + W_o(6R) - W_o(5R) \leq W_o(6R)$$

we obtain: If f satisfies $\Delta^2 f \geq 0$ on $A_o(R-1, 6R+1)$, then

$$\sum_{A(2R,5R)} |\nabla \log \Delta f|^2 \leq 64\,\theta'^4 \frac{W_o(6R)}{R^2} \leq 64\,\theta'^4 \frac{V_o(6R)}{R^2}.$$

For Harnack inequality part on annuli, we also have:

Theorem 4.7 (Harnack Inequality on Annuli)**.** *Let (G,ω) be a connected grid with*

$$\overline{\omega} = \sup_{e\in E} \frac{1}{\omega(e)} < \infty.$$

Fix $o \in X$ and integers s, R with $0 < s \leq R$. Assume G satisfies $[C(o,s,6R)]$. Then for any f satisfying $\Delta^2 f \geq 0$ on $A_o(R-1, 6R+1)$,

$$\sup_A \Delta f \leq \exp\left(32 s^2 \overline{\omega}\, \theta'^2 \frac{V(6R)}{R^2} \right) \inf_A \Delta f,$$

where

$$A = A_o(3R{-}1, 4R{+}1), \qquad \theta' = \sup\left\{ \log \tfrac{\Delta f(x)}{\Delta f(y)} : x, y \in A_o(R{-}1, 6R{+}1),\ x \sim y \right\}.$$

If, in addition,

$$Q = \sup_{r\geq 1} \frac{V_o(r)}{r^2} < \infty,$$

then

$$\sup_A \Delta f \leq C \inf_A \Delta f, \qquad C = \exp\left(\frac{8}{9} s^2 \overline{\omega}\, \theta'^2 Q \right).$$

5 Summary

We first introduced the concepts of irreducibility and recurrence of Markov chains, together with their intrinsic relationship with motions on graphs. Hilbert spaces associated with relevant functionals were defined, along with operators such as

$$\Delta,\ D,\ D^*,\ \nabla,\ L,$$

and the Dirichlet norm. Furthermore, we discussed the connection between superharmonic functions and the Laplacian:

$$f \text{ is superharmonic} \iff \Delta f \geq 0.$$

On manifolds, we derived the gradient estimate

$$\int_{B(r_0)} |\nabla \log \Delta f|^2 \leq -8\theta'^2 \int_{B(r_0)} \frac{\Delta^2 f}{\Delta f} + 64\theta'^4 \left(\sum_{i=0}^{l-1} \frac{(r_{i+1}-r_i)^2}{W(r_{i+1})-W(r_i)} \right)^{-1}.$$

By applying similar techniques to graphs (G, X), we obtained the discrete analogue

$$\sum_{B_0(r_0)} |\nabla \log \Delta f|^2 \leq 64\theta'^4 \left(\sum_{i=0}^{l-1} \frac{(r_{i+1}-r_i)^2}{W(r_{i+1})-W(r_i)} \right)^{-1}.$$

In Section 4, this inequality was extended to annular regions:

$$\sum_{A(s,t)} |\nabla \log \Delta f|^2 \leq 64\theta'^4 \left(\sum_{i=0}^{l-1} \frac{(n_{i+1}-n_i)^2}{U(n_{i+1})-U(n_i)} \right)^{-1}, \qquad A_o(r,R) = B_o(R) \backslash B_o(r).$$

Regarding parabolicity, we established the equivalence among the following four properties:

- recurrence of the associated Markov chain,
- parabolicity of the graph,
- the condition $f \equiv 1 \in D^0(N)$,
- every nonnegative superharmonic function is constant.

These equivalences rely on the minimal element lemma in Hilbert spaces, the "energy" associated with motion on grids, and variational methods. Moreover, we proved that a Markov chain is recurrent if and only if every excessive measure is a multiple of some invariant measure, a result whose proof involves the well-ordering theorem.

An important criterion for recurrence is the Nash–Williams condition:

$$\sum_{i=1}^{\infty} \frac{1}{\omega'(i-1,i)} = \infty.$$

Using the gradient estimate from Section 2, we also showed that

$$\sum_{i=0}^{\infty} \frac{(r_{i+1}-r_i)^2}{W(r_{i+1})-W(r_i)} = +\infty \quad \Longleftrightarrow \quad \inf E_2(f) = 0,$$

which is precisely the condition of bi-parabolicity.

Finally, in Chapter 4, we derived Harnack-type inequalities of the form

$$\sup_{B_0(R+1)} \Delta f \leq C \inf_{B_0(R+1)} \Delta f,$$

and demonstrated that analogous estimates also hold on annular regions:

$$\sup_{A} \Delta f \leq C' \inf_{A} \Delta f,$$

where C and C' are constants depending only on the geometry of the corresponding grid region.

References

[1] S. Y. Cheng and S. T. Yau, *Differential equations on Riemannian manifolds*, Communications on Pure and Applied Mathematics **28** (1975), 333–354.

[2] L. Saloff-Coste, *Inequalities for p-superharmonic functions on networks*, Milan Journal of Mathematics **65** (1995), 139–158.

[3] W. Woess, *Random Walks on Infinite Graphs and Groups*, Cambridge Tracts in Mathematics, vol. 138, Cambridge University Press, 2000.

[4] G. F. Lawler, *Estimates for differences and Harnack inequality for difference operators arising from random walks with symmetric, spatially inhomogeneous increments*, Proceedings of the London Mathematical Society **63** (1991), 449–672.

[5] S. T. Yau, *Harmonic functions on complete Riemannian manifolds*, Communications on Pure and Applied Mathematics **28** (1975), 201–228.

[6] P. M. Soardi and M. Yamasaki, *Classification of infinite networks and its application*, Circuits Systems Signal Process **12** (1993), 133–149.

[7] K. L. Chung and W. H. J. Fuchs, *On the distribution of sums of random variables with copula-induced dependence*, Insurance: Mathematics and Economics **58** (2014), 27–44.

[8] C. St. J. A. Nash-Williams, *Random walks and electric currents in networks*, Mathematical Proceedings of the Cambridge Philosophical Society **55** (1959), 181–194.

[9] F. Spitzer, *Principles of Random Walk*, Graduate Texts in Mathematics, vol. 34, Springer-Verlag, 1976.

[10] K. L. Chung, *Markov Chains with Stationary Transition Probabilities*, Grundlehren der Mathematischen Wissenschaften, vol. 104, Springer-Verlag, 1960.

[11] W. Rudin, *Real and Complex Analysis*, 3rd ed., McGraw–Hill, 1987.

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